(ii) Find x_i the lower class boundary

(iii) Find $z_i = \frac{x_i - \mu}{\sigma}$

(iv) Find Z_i (Zi) = $\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{z} e^{-\frac{z^2}{2}} dz$

(v) Find $\triangle \phi$ (Z_i) = ϕ (Z_{i+1}) - ϕ (Z_i)

(vi) Find expected frequency E_i = = $N \triangle \phi(Z_i)$

Example 10.36

Find expected frequencies for the following data, if its calculated mean and standard deviation are 79.945 and 5.545.

Class	60-65	65-70	70-75	75-80	80-85	85-90	90-95	95-100
Frequency	3	21	150	335	326	135	26	4

Solution:

Given $\mu = 79.945, \sigma = 5.545$, and $N = 1000$

Hence the equation of Normal curve fitted to the data is

$$f(x) = \frac{1}{5.545\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-79.945}{5.545}\right)^2}$$

Theoretical Normal frequencies can be obtained as follows:

Class	Lower Class boundry (X_i)	$z_i = \frac{X_i - \mu}{\sigma}$	ϕ (Z_i) = $\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{z} e^{-\frac{z^2}{2}} dz$	$\triangle\phi$ (Z_i) = ϕ (Z_{i+1}) – ϕ (Z_i)	Expected frequencies $N \triangle \phi(Z_i)$
Below 60	- ∞	- ∞	0	0.0001	0
60 - 65	60	-3.59693	0.001	0.0035	3.5 ≈ 4
65 - 70	65	-2.69522	0.0036	0.0331	33.1 ≈ 33
70 - 75	70	-1.79351	0.0367	0.15	150
75 - 80	75	-0.89179	0.1867	0.3173	317.3 ≈ 317
80 - 85	80	0.009919	0.504	0.3146	314.6 ≈ 315
85 - 90	85	0.911632	0.8186	0.1463	146.3 ≈ 146
90 - 95	90	1.813345	0.9649	0.0318	31.8 ≈ 32
95 – 100	95	2.715059	0.9967	0.0032	3.2 ≈ 3
100 and Over	100	3.616772	0.9999		